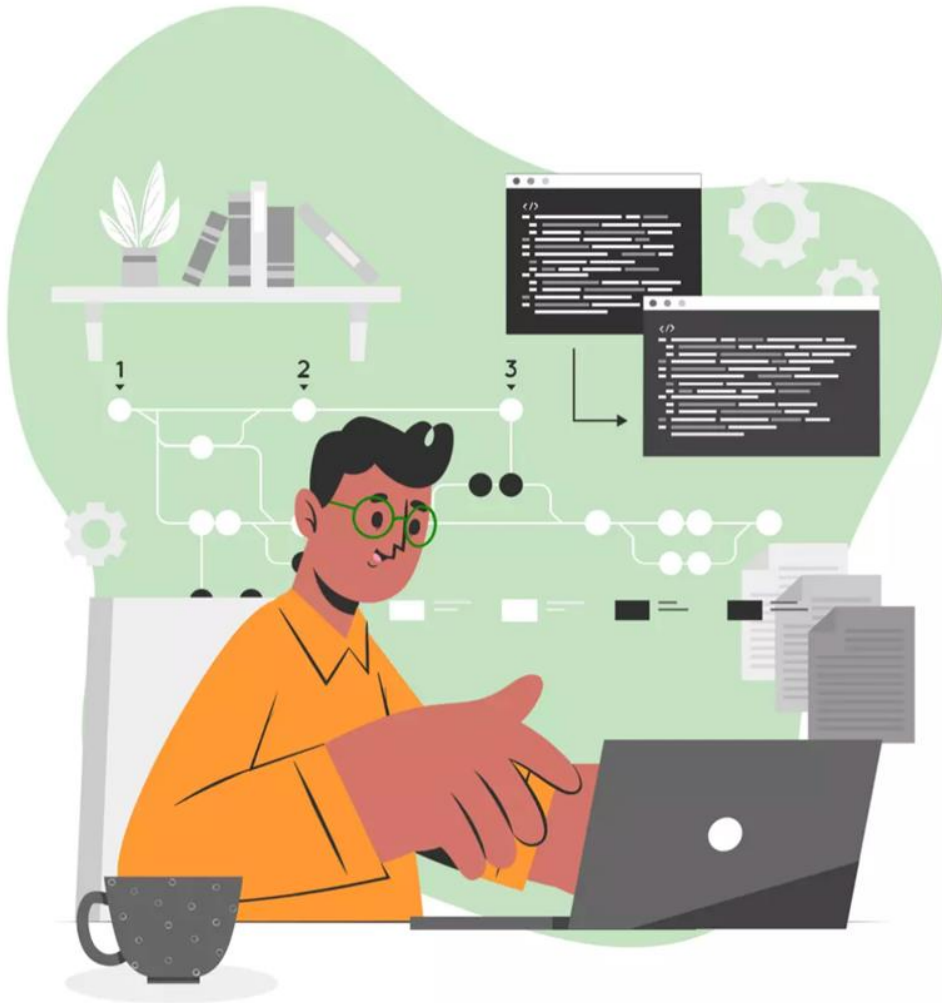




Data Science Workflow

By: Dr. Hassan Naderi

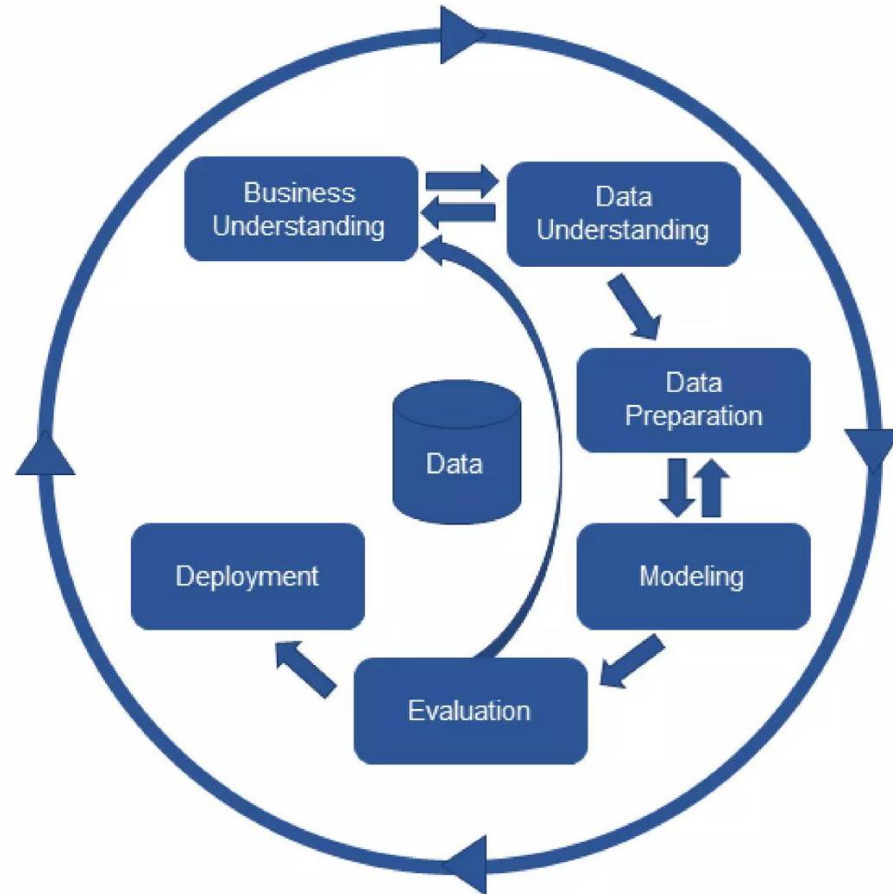
...



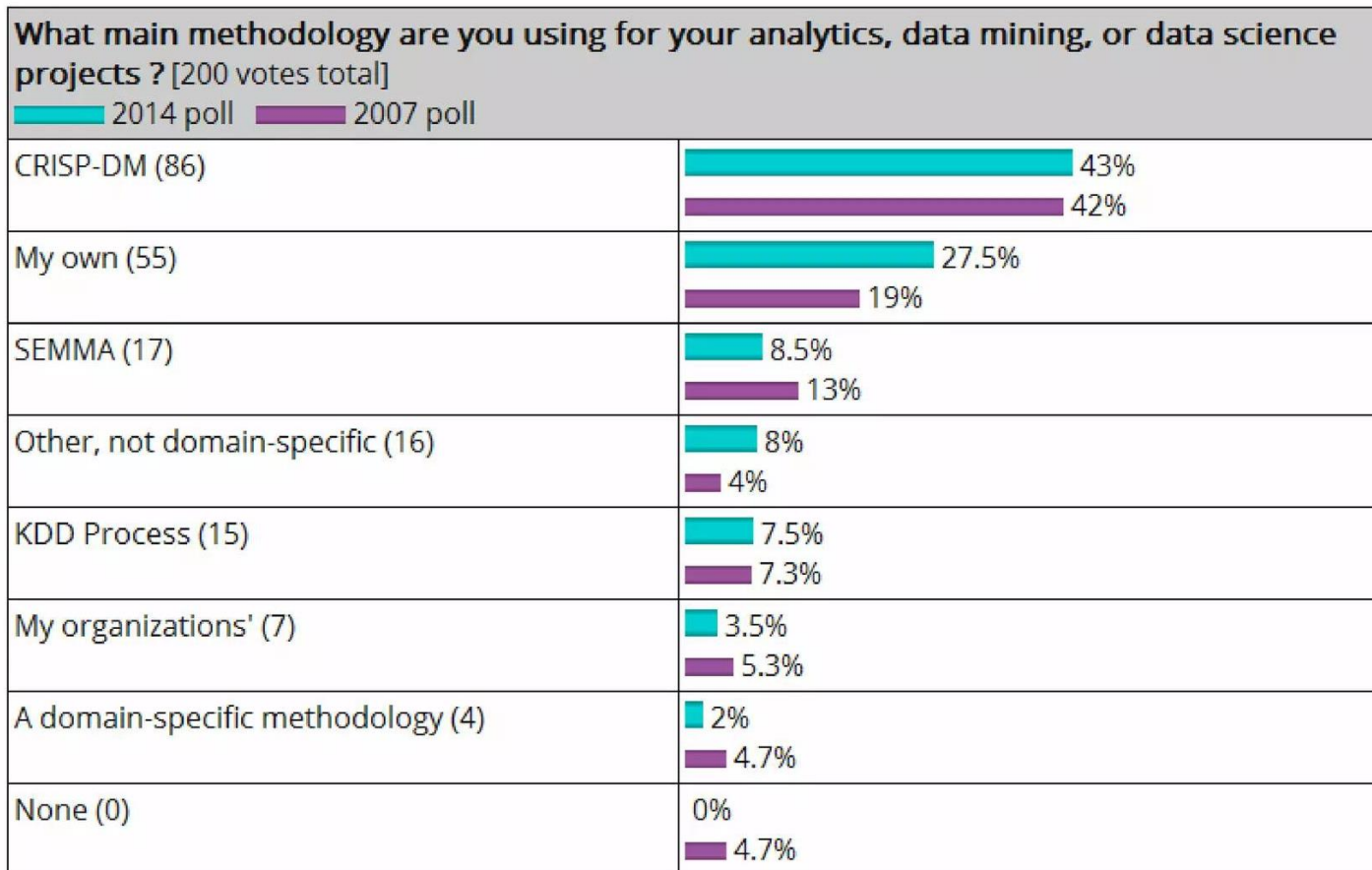
Different Data Science Workflows



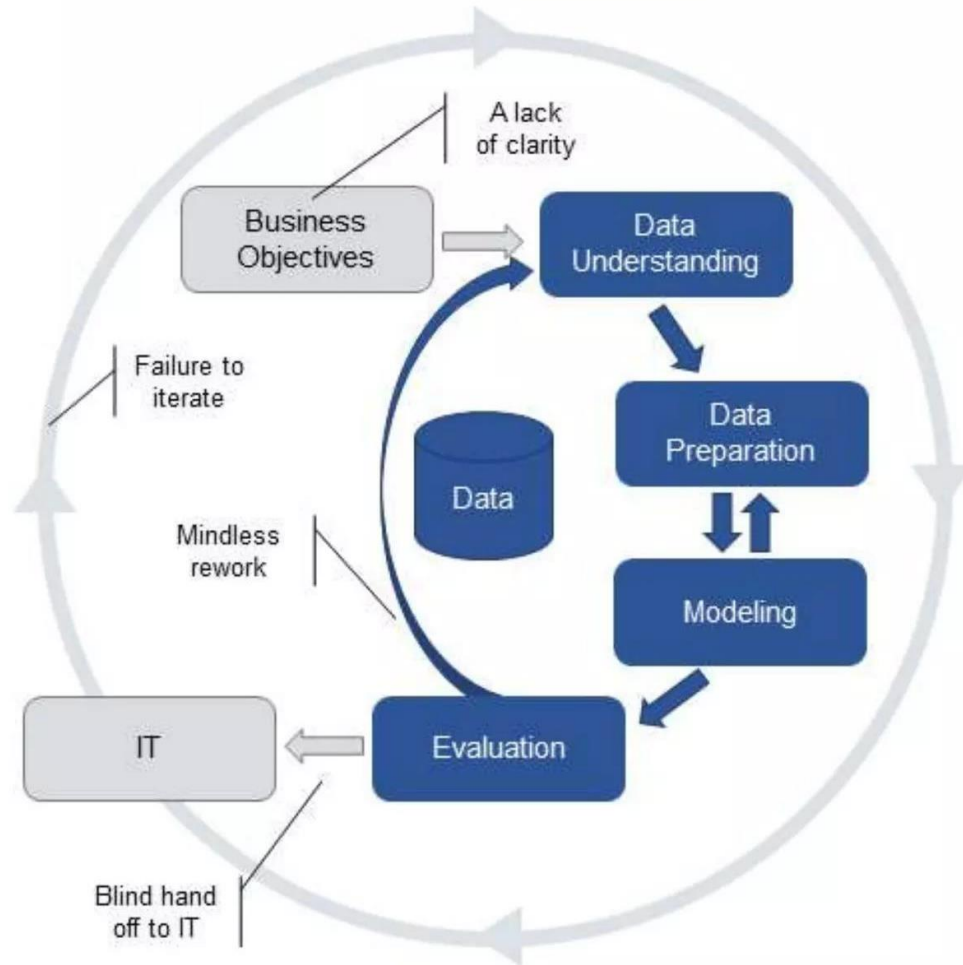
Cross Industry Standard Process for Data Mining (CRISP-DM)



CRISP-DM was the most popular methodology



However, the most popular is not always the best...



- Created in the 1990's
- No longer maintained or updated with new technologies such as Big Data
- Not really focus on the team aspect of working in Advanced Analytics projects

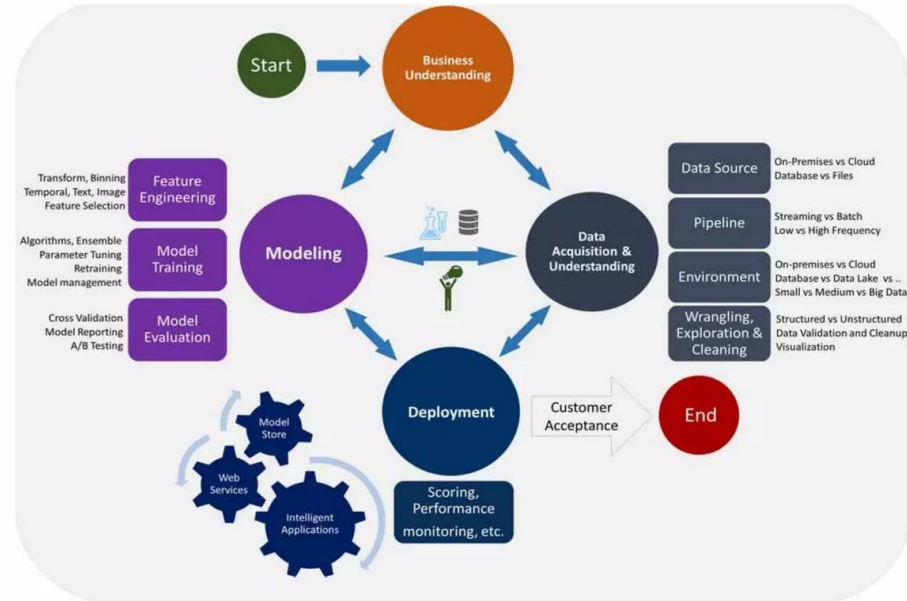
The new winner is...

Team Data Science Process

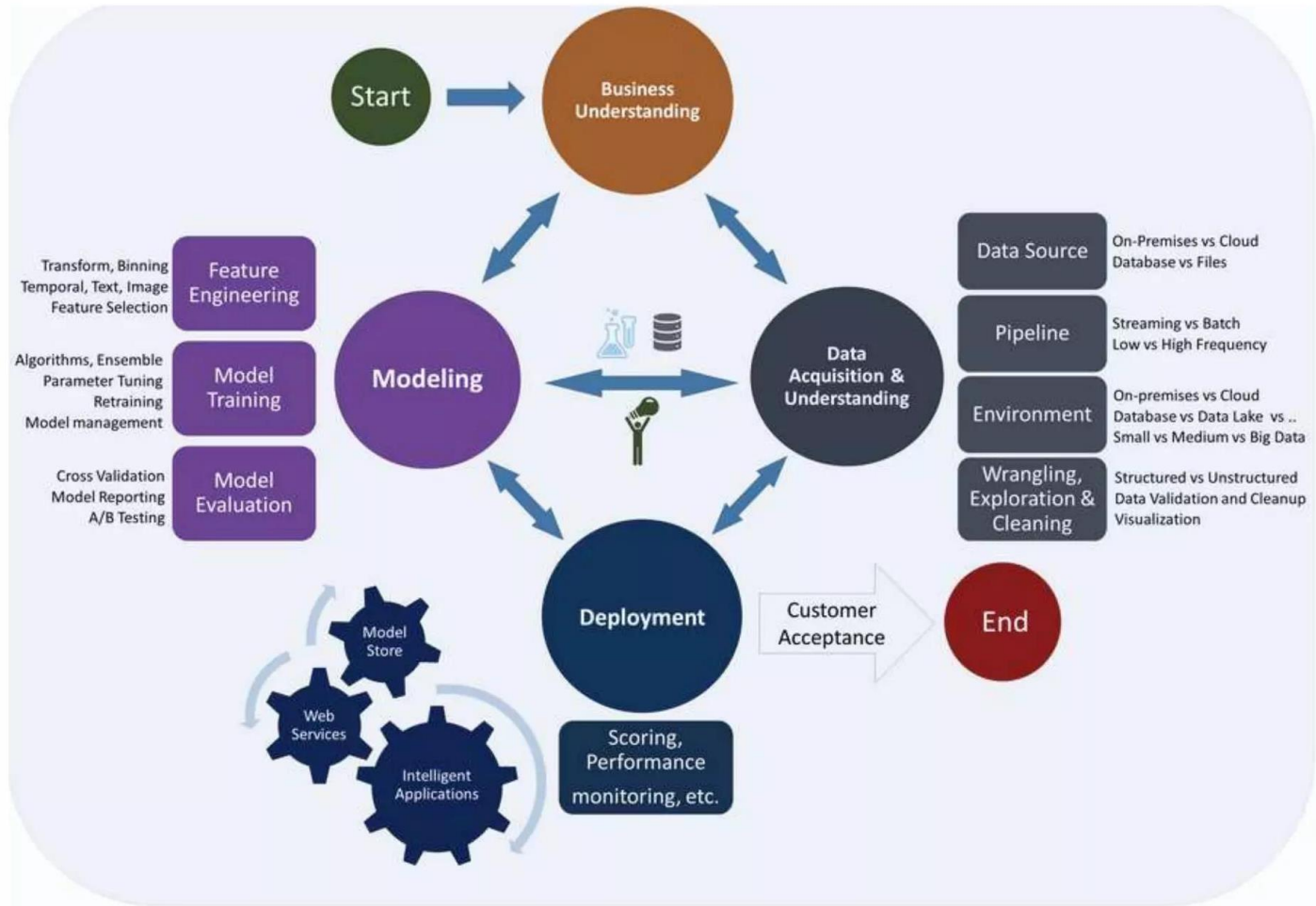
an agile, iterative, data science methodology to improve collaboration and team learning.

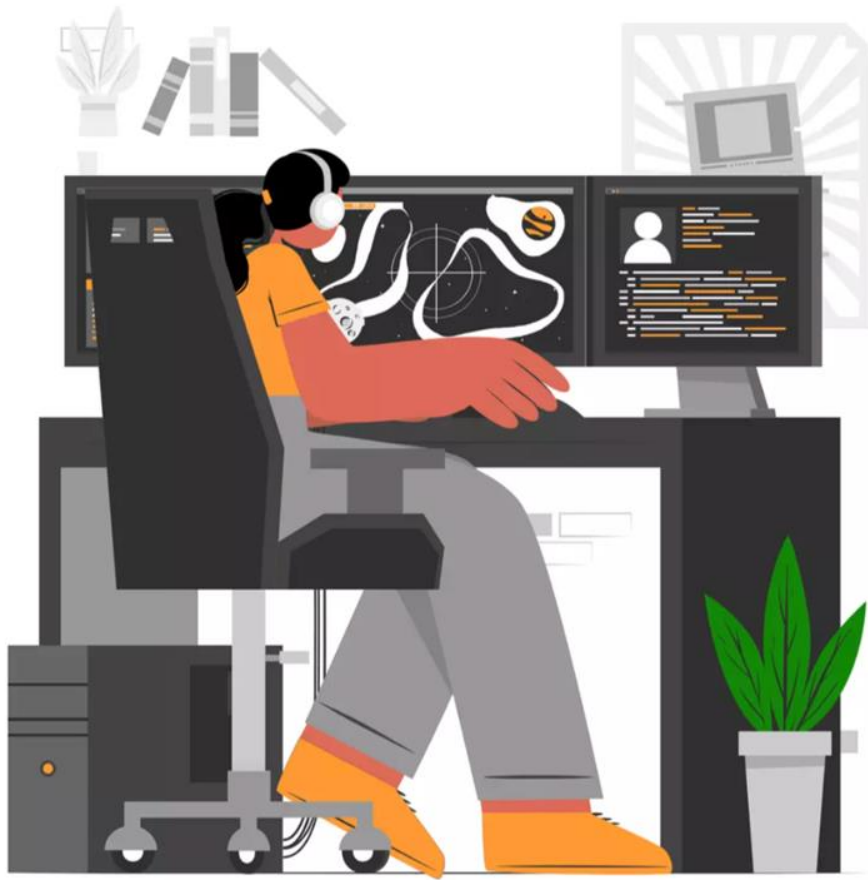
It has the following components:

- **data science lifecycle**
- **standard project structure**
- shared and distributed analytics infrastructure
- productivity tools and utilities



Data Science Lifecycle





Business Understanding

Define the Problem

- 
- **Problem Domain Understanding** Gaining insight into the industry or field in which the problem resides. This involves understanding the nuances, challenges, and intricacies of that domain.
 - **Objective Identification** Clearly outline the objectives and goals of the analysis. This could be predicting customer behavior, optimizing resource allocation, enhancing product performance, or any other measurable outcome.
 - **Actionable Statement Framing** Converting the problem into a well-defined, actionable statement. This statement should articulate the problem's essence, making it understandable and aligned with the business or project goals.



Data Understanding



Data collection & Data Source

- **Manual Data Collection**

Surveys and Questionnaires, Interviews, Observations

- **Automated Data Collection**

Web Scraping, APIs (Application Programming Interfaces, Sensors/IoT, Devices Logs, Telemetry

- **Public and Open Datasets**

Government Portals (like data.gov European Union Open Data Portal, or local government websites), Academic Datasets (e.g., UCI Machine Learning Repository, Kaggle Datasets), Online Repositories





Data collection cont.

- **Business and Proprietary Data**

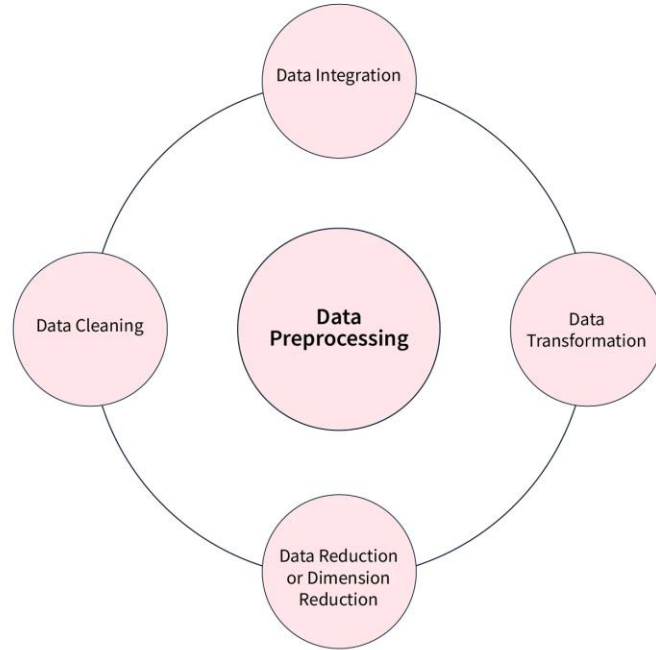
Internal Databases, Customer Feedback, CRM and ERP Systems, Third-Party Data Providers

- **Crowdsourced Data Collecting**

Crowdsourcing Platforms(like Amazon Mechanical Turk, CrowdFlower, or custom applications can help collect input from multiple users), Social Media and Community Contributions



Data Preprocessing

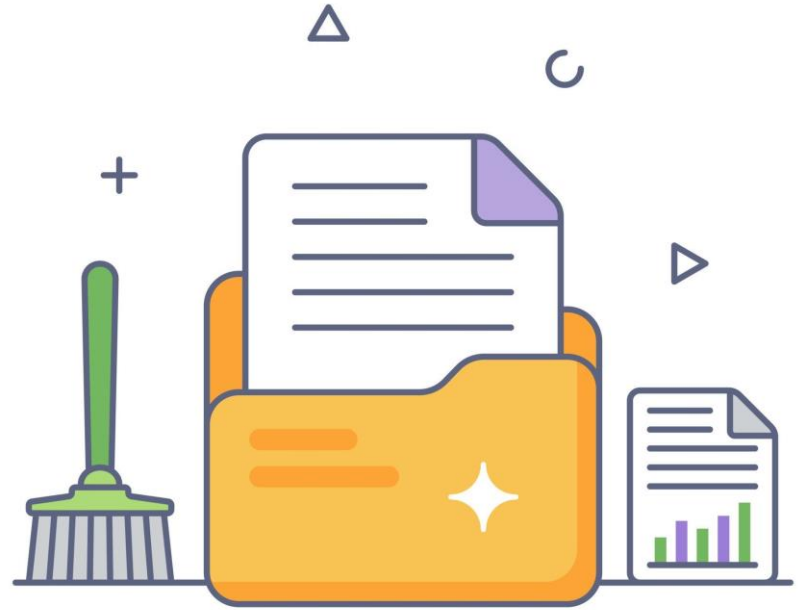


Data Cleaning



Data Cleaning

- Removing Duplicates
- Handling Missing Values
- Filtering Outliers and irrelevant
- Normalizing Data





Data Integration

it can be defined as combining data from multiple sources. A few of the issues to be considered during Data Integration include the following;

- **Entity Identification Problem**

It can be defined as identifying objects/features from multiple databases that correspond to the same entity.

- **Schema Integration**

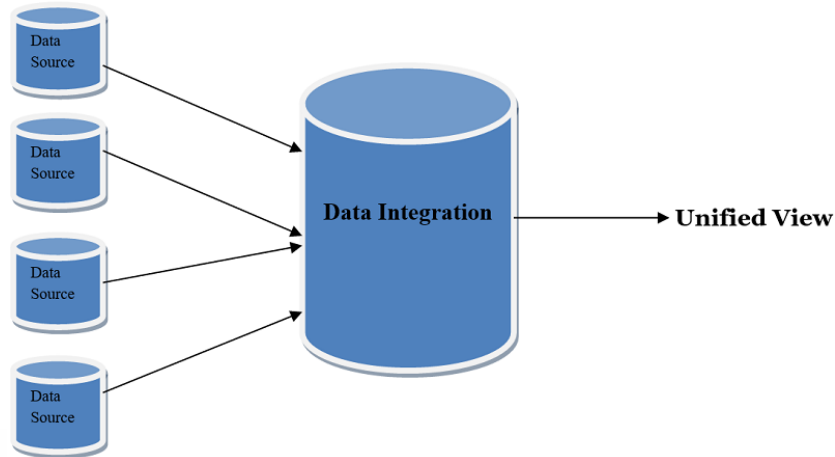
It is used to merge two or more database schema/metadata into a single schema. It essentially takes two or more schema as input and determines a mapping between them. For example, entity type CUSTOMER in one schema may have client in another schema.



Data Integration

- Detecting and Resolving Data Value Concepts

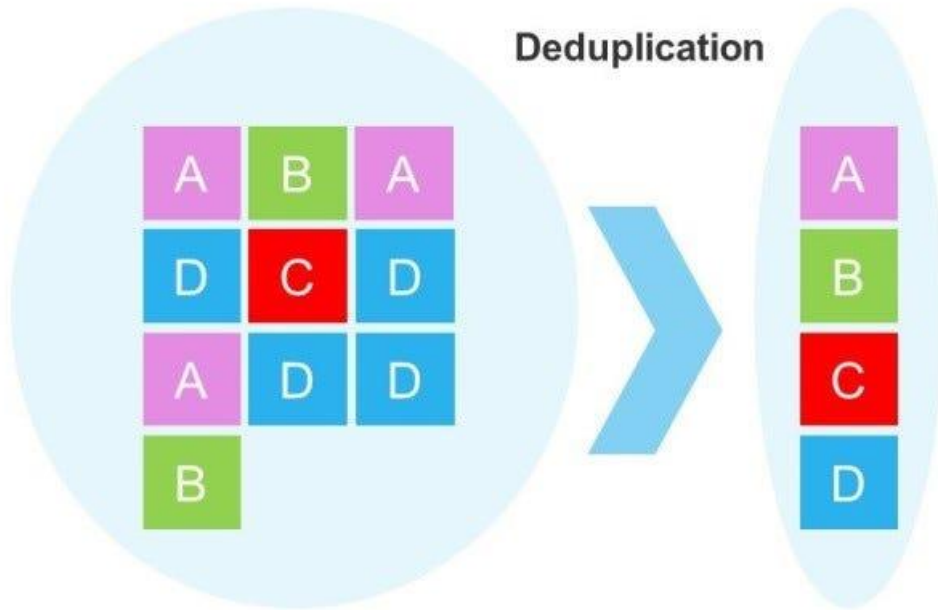
The data can be stored in various ways in different databases, and it needs to be taken care of while integrating them into a single dataset. For example, dates can be stored in various formats such as DD/MM/YYYY, YYYY/MM/DD, or MM/DD/YYYY, etc.



Data Reduction

- 
- **Dimensionality Reduction** - It is the process of reducing the number of features in the input dataset. It can be performed in various ways, such as selecting features with the highest importance, Principal Component Analysis (PCA), etc.
 - **Numerosity Reduction** - In this method, various techniques can be applied to reduce the volume of data by choosing alternative smaller representations of the data. For example, a variable can be approximated by a regression model, and instead of storing the entire variable, we can store the regression model to approximate it.
 - **Data Compression** - In this method, data is compressed. Data Compression can be lossless or lossy depending on whether the information is lost or not during compression.

Data Reduction



Compression



Data Transformation

- 
- **Smoothing** Data Smoothing is used to remove noise in the dataset, and it helps identify important features and detect patterns. Therefore, it can help in predicting trends or future events.
 - **Aggregation** Data Aggregation is the process of transforming large volumes of data into an organized and summarized format that is more understandable and comprehensive. For example, a company may look at monthly sales data of a product instead of raw sales data to understand its performance better and forecast future sales.

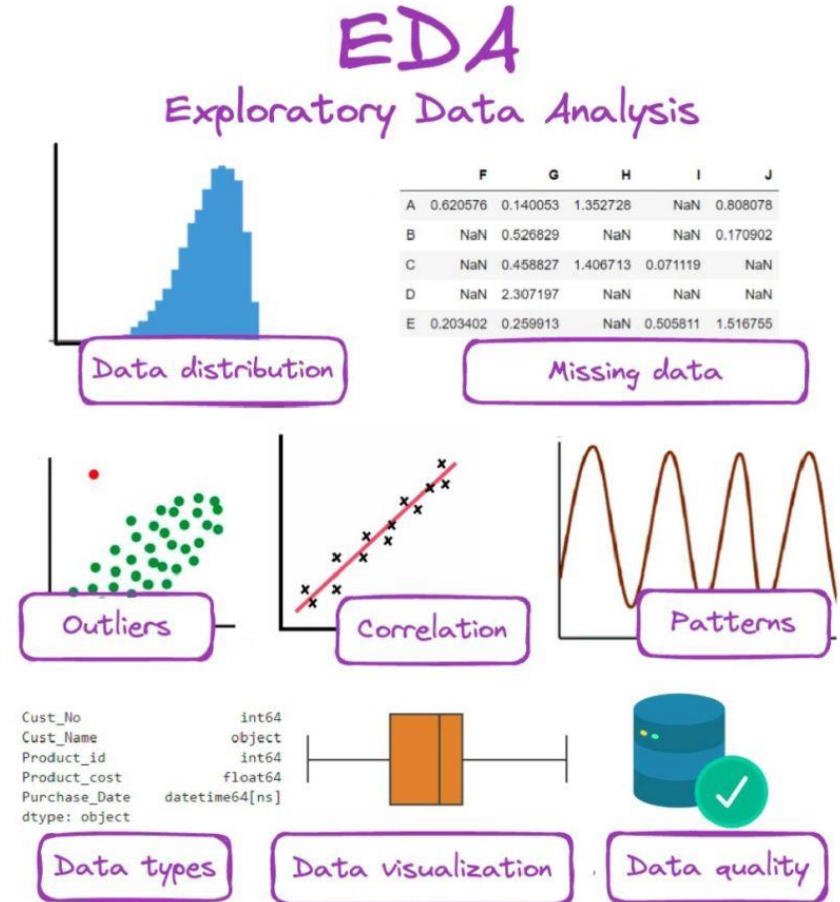
Data Transformation

- 
- **Discretization** - Data Discretization is a process of converting numerical or continuous variables into a set of intervals/bins. This makes data easier to analyze. For example, the age features can be converted into various intervals such as (0-10, 11-20, ..) or (child, young, ...).
 - **Normalization** - Data Normalization is a process of converting a numeric variable into a specified range such as $[-1,1]$, $[0,1]$, etc. A few of the most common approaches to performing normalization are **Min-Max Normalization**, **Data Standardization** or **Data Scaling**, etc.

Analyze Data

With a clean dataset, the focus shifts towards extracting insights and patterns. Analyzing the data involves:

- **Exploratory Data Analysis (EDA)**
Examining the data visually and statistically to understand its characteristics, distributions, correlations, and outliers.



Analyze Data



- **Feature Engineering** Selecting, transforming, or creating features that best represent the underlying patterns in the data.
- **Model Building and Evaluation** Applying suitable algorithms and methodologies to build models, followed by rigorous evaluation to ensure their effectiveness.

Interpret Results

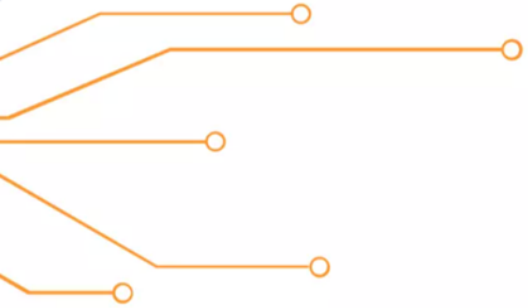


Interpreting the analyzed data is crucial to extract actionable insights and communicate them effectively. Key actions in this step include:

Deriving Meaningful Conclusions Translating the analysis results into meaningful and actionable insights.

Contextual Understanding Relating the findings to the original problem's context to understand their significance and impact.

Effective Communication Present the insights in a clear, understandable manner using visualization tools, reports, or presentations. This aids in communicating the results to stakeholders, enabling informed decision-making.



The End

...

